Do-It-Yourself Assistive Technology

CMU 17-332/732 05-332/632: Introduction to Accessibility and Assistive Technology



Meet Your Presenter

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Assistive Technology Program Manager

- Certified Assistive Technology Professional
- MS in Rehabilitation Science, BS in Mechanical Engineering
- Areas of focus: cellphone, tablet and computer access



Introduction – Why DIY?

- Disability does not have a one size fits all solution – *Universal design can only go so far*
- Assistive technology products can be expensive or difficult to acquire
- A problem may be so unique that it may never get a commercialized solution or product

Outline

- Design Process
- 3D Printing AT
- Switch Adapting & Soldering
- Other DIY AT
- Breakout Activities

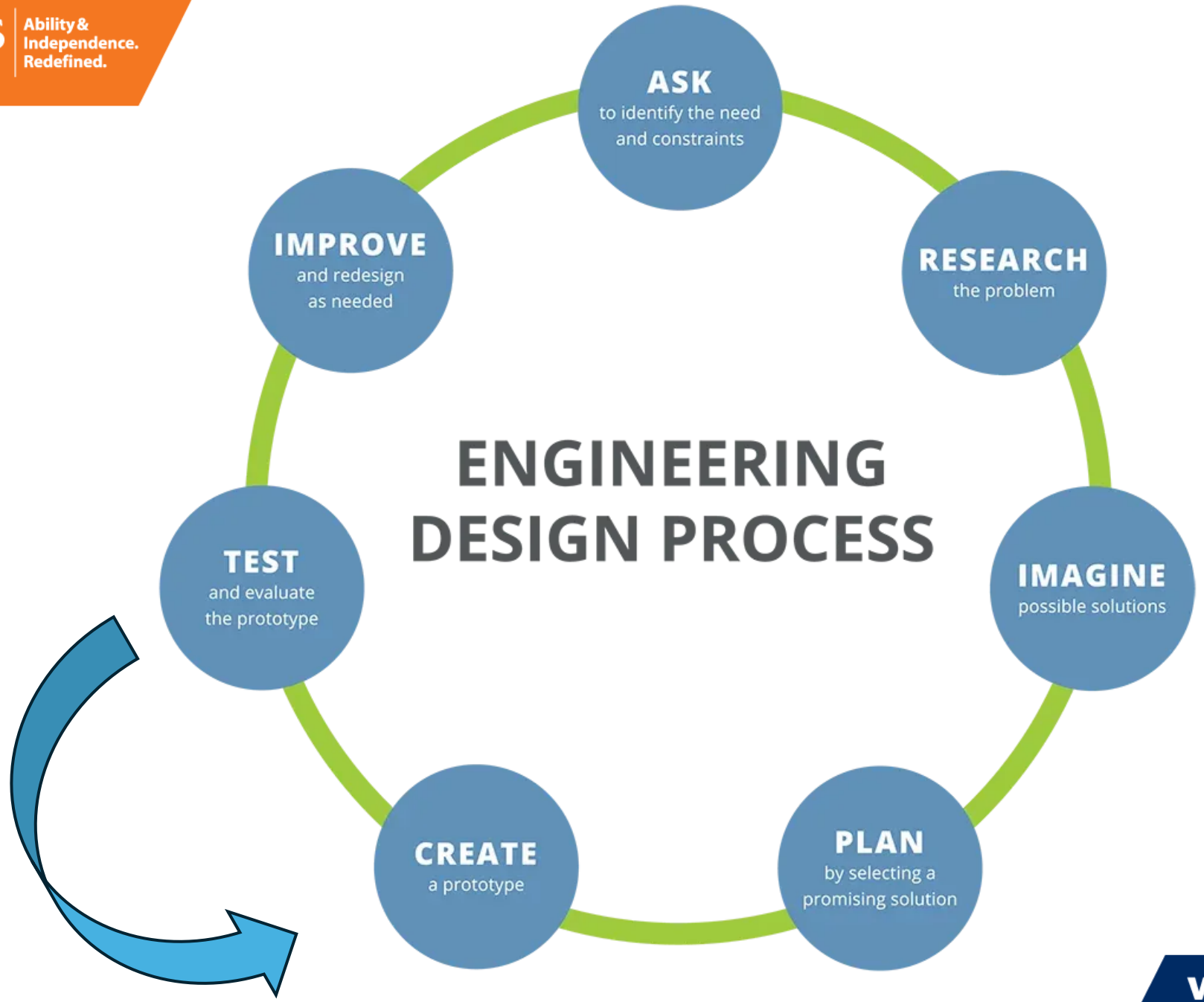


What is Assistive Technology (AT)?

Assistive Technology Device

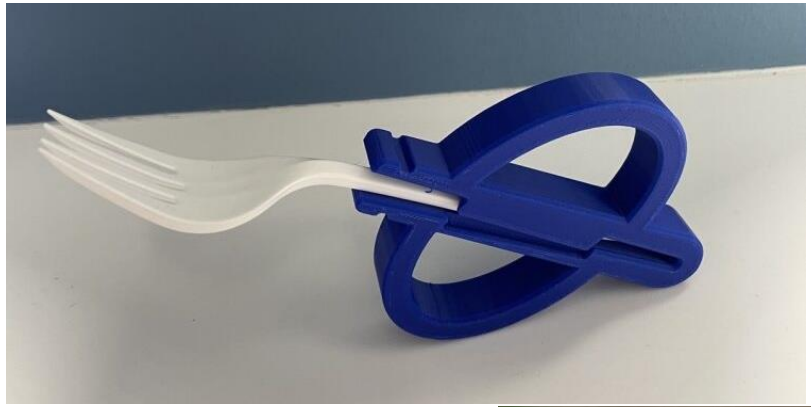
- any item, piece of equipment, or product system, whether acquired commercially, modified, or customized, that is used to increase, maintain, or improve functional capabilities of individuals with disabilities.

ENGINEERING DESIGN PROCESS



3D Printed AT

- Customized solutions
- Quick turnaround times
- Low Cost
- 3D printers are accessible to a lot of individuals
















TV POWER
CBL POWER


TV POWER
CBL POWER
0
VOL
VOL
FAV
VOD
CH
CH
GUIDE
MENU
SEL/OK

3D Printed AT

Barriers

- Liability issues with 3D printed devices
- Durability issues
- Some materials aren’t food safe
- Environmental concerns

Maker Community

Assistive Devices

Build open-source assistive devices for yourself or request a volunteer to make it for you.
How Does It Work?

Search

Search

Viewing 1-18 of 213 Devices

Sort By

Recently Added

Filters

Device Type

- AAC / Keyguard (14)
- Adapted Toy (20)
- Aid for Daily Living (59)
- Assistive Switch (23)
- Device Kit (3)
- Digital Accessory (7)
- GAME Accessory (16)
- Joystick / Mouse (9)
- Low Vision Aid (10)
- Mounting (22)
- Reading / Writing Aid (12)
- Switch Accessory (10)
- Switch Interface (5)

Disability Type



Adaptive Utensil Handles for IKEA IDENTITET



Customizable - Laser-Cut Keyguard for Grid-based-Free-form- and Hybrid AAC Apps on Tablets



Musical Grasping Training Aid

Makers Making Change
A Neil Squire Program

GRiP

Tikkun Olam Makers

Switch Adapted Toys



Soldering Projects

AT for Adaptive Gaming

- Custom peripherals for existing accessible controllers

AT for AAC







Other DIY AT – Instamorph







Other DIY AT – Loc-Line





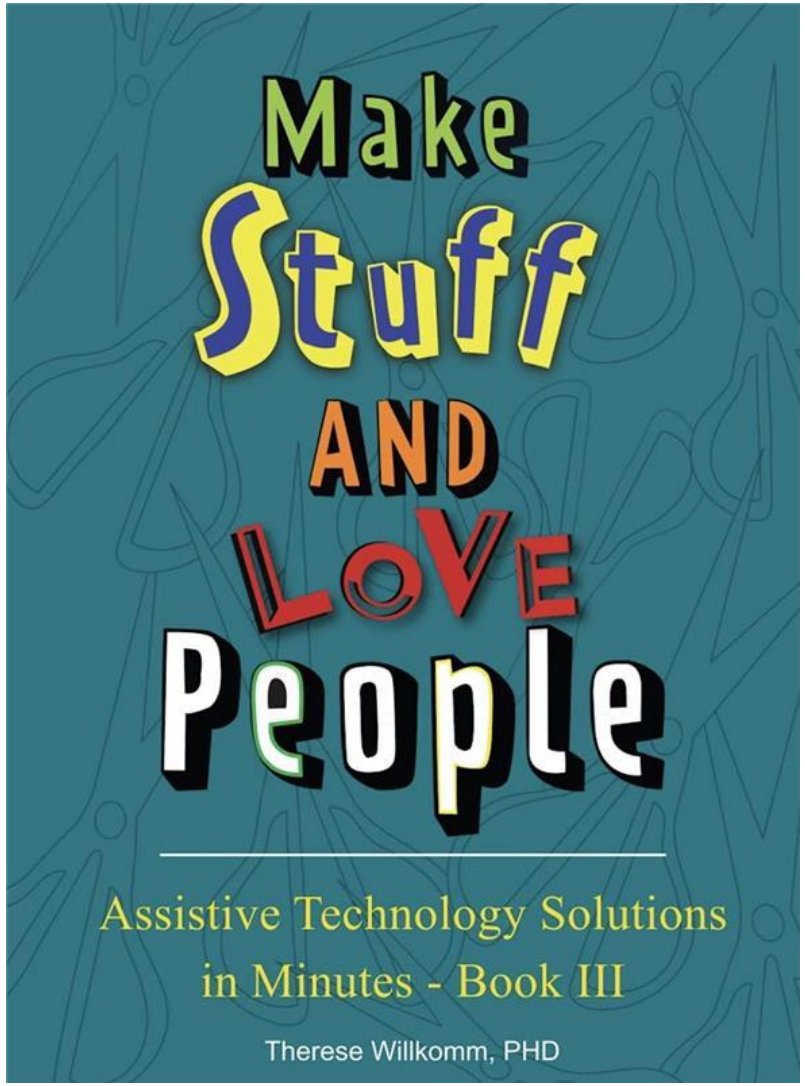
Other DIY AT – Corrugated Plastic





DIY AT Resources

- Assistive Technology Solutions in Minutes Book III - Make Stuff and Love People: Therese Willkomm: 9780989326216: Amazon.com: Books



Staying Involved

- RESNA Student Design Challenge 2027
- Switch Chapters – Switched Adapted Toys
- Inglis: Posts | LinkedIn

Questions?

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Ability & Independence. Redefined.

Breakout Activities